

MACHINE LEARNING

PYTHON 07

SET PRACTICE QUESTION 01

ABHAY RAJ

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Q. write a program to create a set and add a new element to it, remove an element using `remove()`, and discard element using `discard()`.

```
54_sets.py > ...
1  s= {"apple","banana","cherry"}
2  print(s)
3  s.add("guava")
4  print(s)
5
6  #remove
7
8  s.remove("apple") #removing not available element will give error
9
10 print(s)
11
12
13 print(s)
14
15 #discard
16
17 s.discard("gua") #discarding not available element will not give error
18 print(s)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS D:\ML\assignments> & "C:\Program Files\Python314\python.exe" d:/ML/assignments/54_sets.py
{'banana', 'apple', 'cherry'}
{'banana', 'apple', 'guava', 'cherry'}
{'banana', 'guava', 'cherry'}
{'banana', 'guava', 'cherry'}
{'banana', 'guava', 'cherry'}
○ PS D:\ML\assignments>
```

Q. write a program to remove all elements from set using `clear()`.

```
55_sets_clear.py > ...
1  colors = {"red", "green", "blue"}
2
3  (variable) colors: set[str]
4  colors.clear()
5  print(colors)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS D:\ML\assignments> & "C:\Program Files\Python314\python.exe" d:/ML/assignments/55_sets_clear.py
{'red', 'blue', 'green'}
○ PS D:\ML\assignments> |
```

Q. write a python program to determine how many elements are in a set without using built in len() function.

```
56_sets_without_using_len.py > ...
1  animals={"cat","dog","bird","fish"}
2
3
4  count=0
5  for i in animals:
6
7      count+=1
8
9  print("length of the set is:", count)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS D:\ML\assignments> & "C:\Program Files\Python314\python.exe" d:/ML/assignments/56_sets_without_using_len.py
length of the set is: 4
○ PS D:\ML\assignments> |
```